HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Java Spring Boot** REST APIs for inter-service communication, handling approximately **1,500 requests per minute**.
- Implemented robust data persistence layers using **Hibernate** and **JPA**, managing complex entity relationships and ensuring data integrity across relational databases.
- Designed and coded object-oriented solutions in **Java**, applying design patterns and conducting comprehensive unit testing with **JUnit** and **Mockito** to achieve high code quality.
- Investigated complex technical challenges to identify the underlying causes of defects and yield loss, applying critical thinking and data analysis to pinpoint root causes.
- Collaborated closely with process engineering, equipment engineering, and manufacturing teams to resolve defect issues, implementing robust fixes to prevent recurrence.
- Utilized **ACE** for defect-yield correlation, identifying top Pareto items impacting yield and developing creative solutions for improvement.
- Containerized applications using **Docker** and managed deployments to **AWS EC2** instances, configuring basic resource limits for predictable rollouts.

SKILLS

Languages – Java, Python, SQL, R, PL/SQL

Frameworks/Libraries – Spring Boot, Hibernate, JPA, Spring Framework, pandas, NumPy, scikit-learn, FastAPI, Flask

Databases – PostgreSQL, MySQL, Oracle, SQL Server

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

Analytics/BI – Power BI, Tableau, DAX, Power Query, SAS, JMP, ACE, Klarity Defect, Statistical Process Control (SPC)

Methodologies – Agile, Scrum, Root Cause Analysis, Design of Experiments (DoE), Change Management

PROJECTS

Automated Defect Analysis and Reporting Pipeline (HCL Technologies)

- Developed a **Python**-based data pipeline to ingest and process defect data from multiple sources, transforming raw logs into structured datasets for analysis.
- Engineered **SQL** scripts to extract and aggregate data from relational databases (**PostgreSQL**, **Oracle**) for generating daily operational reports.
- Implemented automated report generation using **pandas**, summarizing key defect trends and identifying Pareto-dominant issues for the operations team.

Yield Improvement Initiative for Enterprise Software Module (HCL Technologies)

- Applied **Statistical Process Control (SPC)** techniques to monitor inline defect charts, identifying shifts and responding to excursions to maintain correct control and spec limits.
- Collaborated with cross-functional teams including quality assurance and business analysts to analyze defect patterns and gather requirements for yield enhancement.
- Documented findings and proposed process improvements, contributing to a targeted reduction in defect density for a critical software module.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)